

ELSE 2500

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Topic: Conditional PDF for Continuous RVs.

□ Definition of conditional PDF

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)}, \text{ when } f_X(x) > 0$$

This is the generalization of the conditional PMF.

As before, 2 important things to note are

1) $f_{X,Y}(x,y) = f_{X|Y}(x,y) \cdot f_X(x)$ ← Useful

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 Conditional Marginal

2) If X and Y are independent,

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{f_X(x) f_Y(y)}{f_X(x)}$$

$= f_Y(y)$ ← Knowing X doesn't change the density of Y .

An implication of (1) is that

$$\begin{aligned} P(Y \in A) &= \int_{-\infty}^{+\infty} f_{Y|X}(y \in A | x) f_X(x) dx \\ &= \int_{y \in A} f_Y(y) dy \end{aligned}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f_{XY}(x, y) dx = \int_{-\infty}^{+\infty} f_{Y|X}(y|x) f_X(x) dx$$

Example

Select X uniformly at random from an interval $[0, 1]$. Then select

Y uniformly at random from $[0, X]$.

Q: What is the PDF of Y ?

Information we get from problem statement:

$$f_X(x) = 1, \quad x \in [0, 1]$$

$$f_{Y|X}(y|x) = \frac{1}{x}, \quad y \in [0, x]$$

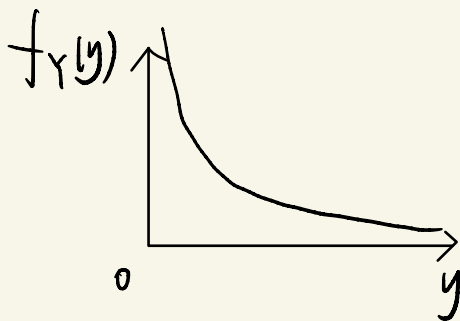
$$\text{So } f_{XY}(x, y) = f_{Y|X}(y|x) \cdot f_X(x) = \frac{1}{x}, \quad 0 \leq y \leq x \leq 1$$

Then we marginalize $f_{XY}(x,y)$ over x ,

$$f_Y(y) = \int_{-\infty}^{+\infty} f_{XY}(x,y) dx$$

$$= \int_y^1 \frac{1}{x} dx = \ln x \Big|_{x=y}^{x=1}$$

$$= \ln 1 - \ln y = -\ln y, \quad 0 \leq y \leq 1$$



That is, small values of Y are more likely than large values.

Example

Say X and Y are the light intensity measures from a laser at time t_1 and t_2 . When the times are close together ($t_1 \approx t_2$), light intensities are correlated with each other. Say that we model X and Y as jointly Gaussian with $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$, correlation coefficient ρ . What is $f_{Y|X}(y|x)$?

To do so, we will use

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} \leftarrow \text{already known}$$

$$\text{Joint PDF } f_{XY}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x^2 - 2\rho xy + y^2)}$$

We need to calculate the marginal PDF

$$f_X(x) = \int_{-\infty}^{+\infty} f_{XY}(x,y) dy$$

$$= \int_{-\infty}^{+\infty} \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x^2 - 2\rho xy + y^2)} dy$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2(1-\rho^2)}(\underbrace{y^2 - 2\rho xy + \rho^2 x^2}_{(y-\rho x)^2} - (1-\rho^2)x^2)} dy$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2(1-\rho^2)}((y-\rho x)^2 - (1-\rho^2)x^2)} dy$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \underbrace{\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2} \frac{(y-\rho x)^2}{(1-\rho^2)}} dy}_{\text{View it as Gaussian with mean } \rho x \text{ and variance } (1-\rho^2)^2}$$

View it as Gaussian with mean ρx and variance $(1-\rho^2)^2$

Integration over a PDF is 1!

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \leftarrow \text{Marginal PDF of } X \text{ is}$$

with mean $\mu_x = 0$, and
variance $\sigma_x^2 = 1$

Now we can get the conditional PDF

$$f_{Y|X}(y|x) = \frac{\frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x^2 - 2\rho xy + y^2)}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}}$$

Gaussian
mean ρx
variance $1 - \rho^2$

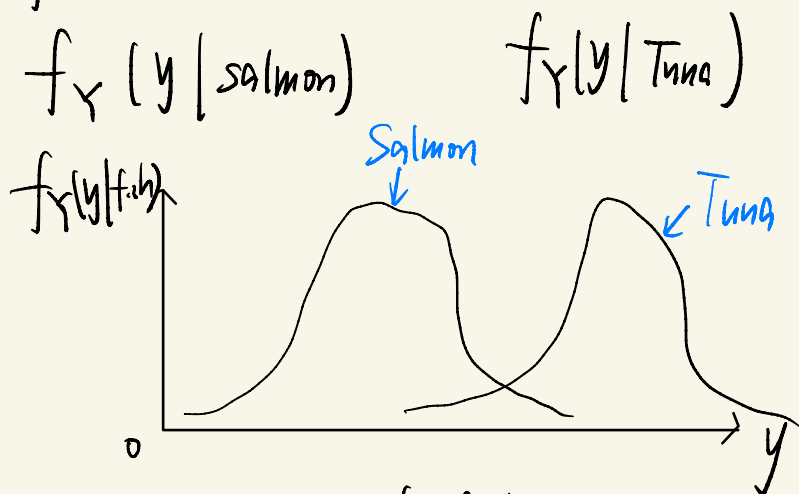
$$\rightarrow = \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2} \frac{(y - \rho x)^2}{1 - \rho^2}}$$

Remark: The conditional PDF of y is centered around ρx . If the observed RV x is $x > 0$, Y is biased to be positive. On the other hand, by the same reasoning, the marginal PDF $f_Y(y)$ is centered around mean 0. Thus, knowing X definitely changes the density of Y .

□ Where the conditional PDF is useful?

In machine learning, we are often given the conditional PDFs (e.g., after observing realizations).

Classic example: We observe the length of a fish we caught in the Hudson river. We want to determine what kind of fish it is?



We want to estimate if fish is Tuna or Salmon based on its length y .

We apply Bayes Rule

$$P(\text{Salmon} | y) = \frac{f_Y(y | \text{salmon}) P(\text{salmon})}{f_Y(y)}$$

Prior

Total probability $\rightarrow$

$$= \frac{f_Y(y | \text{salmon}) P(\text{salmon})}{\sum_{\text{all fish}} f_Y(y | \text{fish type}) \cdot P(\text{fish})}$$

$$(*) = \frac{f_Y(y | \text{salmon}) P(\text{salmon})}{f_Y(y | \text{salmon}) P(\text{salmon}) + f_Y(y | \text{tuna}) P(\text{tuna})}$$

Likelihood

Likewise, we can also have

$$P(\text{tuna} | y) = \frac{f_Y(y | \text{tuna}) P(\text{tuna})}{f_Y(y | \text{salmon}) P(\text{salmon}) + f_Y(y | \text{tuna}) P(\text{tuna})} (*)$$

Posterior

This leads to the famous decision rule
Bayes Decision Rule

Decide Salmon if $P(\text{Salmon} | y) > P(\text{Tuna} | y)$
Tuna if $P(\text{Tuna} | y) > P(\text{Salmon} | y)$

From (*), we can choose Salmon, i.e.,

$$P(\text{Salmon}|y) > P(\text{Tuna}|y) \quad \text{if}$$

$$f_Y(y|\text{Salmon}) P(\text{Salmon}) > f_Y(y|\text{Tuna}) P(\text{Tuna})$$

$$\Leftrightarrow \frac{f_Y(y|\text{Salmon})}{f_Y(y|\text{Tuna})} > \frac{P(\text{Tuna})}{P(\text{Salmon})}$$

Two special cases:

- 1) If $P(\text{Salmon}) = P(\text{Tuna}) = \frac{1}{2}$, then we are just choosing PDF that has a large value at each point of y

$$\rightarrow f_Y(y|\text{Salmon}) > f_Y(y|\text{Tuna})$$

- 2) If $f_Y(y|\text{Salmon})$ and $f_Y(y|\text{Tuna})$ are same, we end up with choosing whatever class has higher prior.

$$\rightarrow P(\text{Salmon}) > P(\text{Tuna})$$

□ Conditional Expectation

$$\text{(Discrete)} \quad E[Y | X = x_k] = \sum_{\substack{y_j \in \text{All values } Y \text{ take} \\ \{y_1, y_2, \dots, y_k\}}} y_j P_{Y|X}(y_j | x_k)$$

$$\text{(Continuous)} \quad E[Y | X = x] = \int_{-\infty}^{+\infty} y f_{Y|X}(y|x) dy$$

Clearly, the conditional expectations are a function of x . We can actually think of this expectation as a new Random variable:

$$g(X) = E[Y | X]$$

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Big case